

UNIVERSITEIT UTRECHT

MASTER THESIS

Addressing the Cold Start Problem in AI-Assisted Systematic Reviews

Getting started when evidence is scarce

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1 Introduction

1.1 The framework: AI-assisted systematic reviews

Researchers and practitioners are continually challenged to base their decisions on the latest scientific evidence (Sackett et al. 1996; Subbiah 2023; Sheridan and Julian 2016; Chalmers, Hedges, and Cooper 2002). To that end, systematic reviews and meta-analyses were developed as rigorous methods of summarizing scientific literature (Grant and Booth 2009; Brignardello-Petersen, Santesso, and Guyatt 2025; Borenstein et al. 2021; Kicinski 2015). However, systematically reviewing large bodies of literature can be time-consuming. In particular, screening titles and abstracts to identify relevant papers is often a major bottleneck limiting the practical applicability of systematic reviews (Nussbaumer-Streit et al. 2021; Bastian, Glasziou, and Chalmers 2010; Borah et al. 2017).

Fortunately, recent advances in machine learning have produced tools that assist researchers in screening papers by prioritizing the most relevant papers for screening first (Van De Schoot et al. 2021; Ouzzani et al. 2016; Ge et al. 2024; Dos Reis et al. 2023). One such tool is *ASReview*, an open-source software package for AI-assisted systematic reviews (Van De Schoot et al. 2021; J. d. Bruin et al. 2025). *ASReview* uses active learning models to iteratively learn from the user’s screening decisions and adaptively prioritize the most relevant papers for screening (see figure 1). AI-assisted screening can significantly reduce the time and effort needed to conduct systematic reviews and may even help identify more relevant papers compared to screening without the help of AI (Teijema et al. 2025; Byrne et al. 2024; Ferdinands et al. 2023; Schoot et al. 2025).

1.2 The problem: the cold start problem

A key challenge limiting the application of AI-assisted systematic reviews is the “cold start” problem (Van De Schoot et al. 2021; Teijema et al. 2025). In the context of AI-assisted systematic reviews, the cold start problem refers to the inability of the active learning model to prioritize relevant papers for screening when it has yet to be trained on any examples of relevant papers¹.

Accordingly, a researcher can be said to be facing a cold start when no examples of relevant papers are available prior to screening. In this case, the researcher may have to start screening papers at random until a relevant paper to train the AI model on is found. This may take a long time, especially when the percentage of relevant papers returned by a systematic search is low (Bastian, Glasziou, and Chalmers 2010; Borah et al. 2017).

Notably, screening papers at random until a relevant paper has been found does not appear to impact the overall performance of AI-assisted systematic reviews (Oude Wolcherink et al. 2023). One possible explanation is that, once a relevant paper is identified, the AI model has already amassed a substantial training set comprising all the papers labeled as irrelevant up to that point, along with the one relevant paper. This allows the AI model to quickly ‘catch up’ compared to starting with AI screening from the very beginning. In other words, a cold start only affects the *starting performance* of AI-assisted systematic reviews.

Thus, the cold start problem is an issue when no prior examples of relevant papers are available and decisions need to be made in a timely manner. This may be the case

¹The cold start problem is a well-known issue in recommender systems and active learning more generally (Panda and Ray 2022; Lika, Kolomvatsos, and Hadjiefthymiades 2014; Bobadilla et al. 2013).

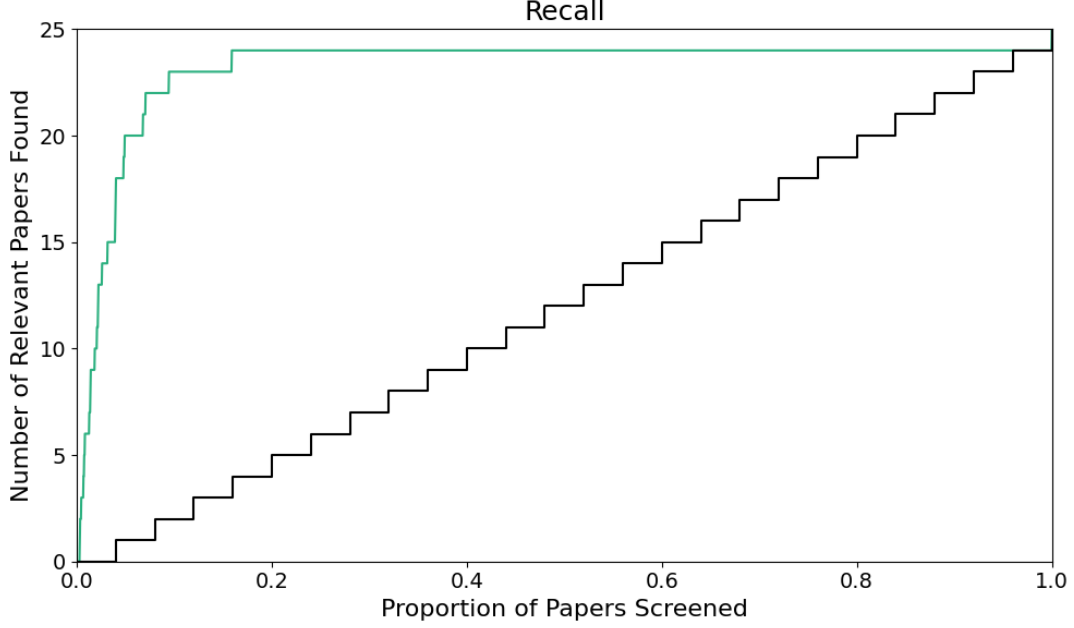


Figure 1: Recall plot of AI-assisted screening and manual screening. The x-axis shows the proportion of papers screened (0–1) and the y-axis shows the number of relevant papers found. The blue line shows a simulated screening of the *Appenzeller-Herzog-2019* dataset with ASReview and the black line shows the expected recall curve when manually screening.

in crisis situations such as COVID-19 (Funtowicz and Ravetz 1994; Evans 2022). In the early days of the pandemic, evidence-based decision-making relied on consulting experts and conducting rapid reviews, which sacrifice rigor for speed (Tricco et al. 2020; Kurzer and Ornston 2023). Therefore, addressing the cold start problem in AI-assisted systematic reviews can improve evidence-based decision-making, particularly in crisis situations where timely evidence synthesis is crucial.

1.3 The proposed solution: eligibility criteria and large language models

A possible solution to the cold start problem is to use the systematic review’s eligibility criteria to initialize the active learning model. The eligibility criteria of a systematic review are typically well-defined and provide a clear description of what constitutes a relevant paper for that review (Cochrane 2026). By using the eligibility criteria as an example of a relevant paper, the active learning model can be trained on this example before screening begins, thus potentially avoiding a cold start. Moreover, it may be possible to further improve starting performance by using large language models (LLMs) to generate synthetic examples of relevant papers based on the eligibility criteria (Riegler et al. 2025). LLMs have been shown to be able to address the cold start problem by leveraging their training data (X. Zhang et al. 2023; W. Zhang et al. 2025; Xia et al. 2025; Bayer and Reuter 2024; Bachmann et al. 2025). LLMs may therefore be able to generate examples that are more similar to actual abstracts of relevant papers than the eligibility criteria by themselves (Steyvers et al. 2025). However, the use LLMs may also add unnecessary steps or even negatively impact starting performance by adding noise

(Gartlehner et al. 2025; Sollini et al. 2025).

1.4 Objectives

Our study has two main objectives:

1. To investigate whether the starting performance of AI-assisted systematic reviews can be improved by initialising the AI-model with synthetic abstracts generated by an LLM, based on a systematic review’s eligibility criteria criteria or by directly using the eligibility criteria criteria as an example of a relevant paper, compared to a cold start.
2. To investigate whether the use of LLM-generated examples improves starting performance beyond that achieved through initialisation using the systematic review’s eligibility criteria criteria directly.

2 Methodology

2.1 Simulation study design

We use ASReview to simulate AI-assisted, title-based screening for systematic reviews². Each simulation run is initialized using one of the four starting conditions described below. To ensure reproducibility, the simulation is replicated across 23 previously screened and labeled systematic reviews from the SYNERGY collection (De Bruin et al. 2023).

The four different types of starting conditions are the *LLM condition*, the *cold-start condition*, the *true-example condition*, and the *criteria-only condition* (for a visual explanation of the four conditions, see results section 3.2).

In the *LLM condition*, the eligibility criteria of the published systematic reviews are provided to OpenAI’s GPT-4o-mini model with the request to generate at least one set of titles and abstracts of a paper relevant to the review at hand³. These sets of titles and abstracts are then given to ASReview along with the published systematic review to screen.

In the *cold-start condition*, no papers are provided to ASReview prior to the start of screening. In the *true-example condition*, one relevant paper is sampled from the set of papers labeled as relevant to the systematic review at hand (with replacement between runs) and provided to ASReview before screening. In the *criteria-only condition*, the eligibility criteria of the systematic review function directly as an example of an abstract of a relevant paper with which to initialise ASReview, rather than using the LLM to generate an example.

Finally, the eligibility criteria of the systematic reviews are manually manipulated in several ways to investigate whether this affects starting performance. Specifically, the

²Two sequential configurations of ASReview are used: the first queries papers at random until at least one relevant and one irrelevant paper have been found and ASReview’s classifier can be trained. Next the U4 configuration is used for the remainder of the screening process. Notably, however, in all conditions except for the cold-start condition, the first configuration is effectively skipped since at least one relevant paper is provided before screening begins and irrelevant papers are plentiful in all systematic reviews. See J. d. Bruin et al. (2025) for a detailed description of the U4 configuration.

³OpenAI’s gpt-4o-mini model is instructed via a DSPY module for dynamic prompt generation in each run (Khattab et al. 2023)

alterations made are: the presence of exclusion criteria, expansion of abbreviations, and expansion of generic terms using MeSH⁴.

The total number of simulation runs is 4 (conditions) x 23 (systematic reviews) x 2 (manipulated vs. unmanipulated eligibility criteria) = 184 runs. FIX!!!

2.2 Analysis plan

Starting performance will be compared between the four conditions visually using a box-plot and statistically using a linear model with bootstrapped SEs. Starting performance is defined as the number of relevant papers successfully retrieved from among the first 100 papers screened. Some runs may have fewer than 100 screened papers because all relevant papers are retrieved and screening is stopped before the 100th screened paper. Accordingly, the total number of papers screened will be included in the model to account for this. The model will also include a dummy variable for dataset to control for the fact that some datasets may be easier to screen than others. To specifically address the two objectives of the study described above, three contrasts will be compared for statistical significance differences in starting performance: (1) LLM vs. cold-start, (2) criteria-only vs. cold-start, and (3) LLM vs. criteria-only. An additional analysis will be conducted to explore the impact of manually manipulating the eligibility criteria.

2.3 Comparing starting performance: recall curves for each condition

3 Results

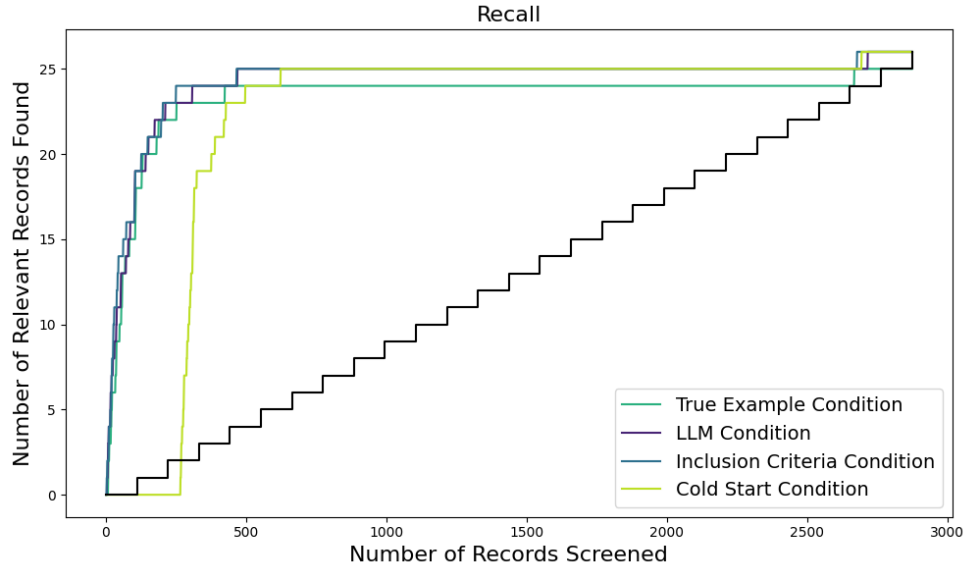
3.1 Assessing starting performance: recall curves

A recall curve is a plot that shows the number of relevant papers retrieved (y-axis) as a function of the number of papers screened (x-axis). The recall curve can be interpreted as follows: the steeper the curve, the better the starting performance, since more relevant papers are retrieved with fewer papers screened. A flat curve indicates poor starting performance, as it suggests that no relevant papers are being retrieved even after screening many papers. Since the focus of this study is on starting performance, which is defined as the number of relevant papers successfully retrieved from among the first 100 papers screened, only the first 100 papers screened are shown in the recall plots in the following sections (i.e., the box in figure ??).

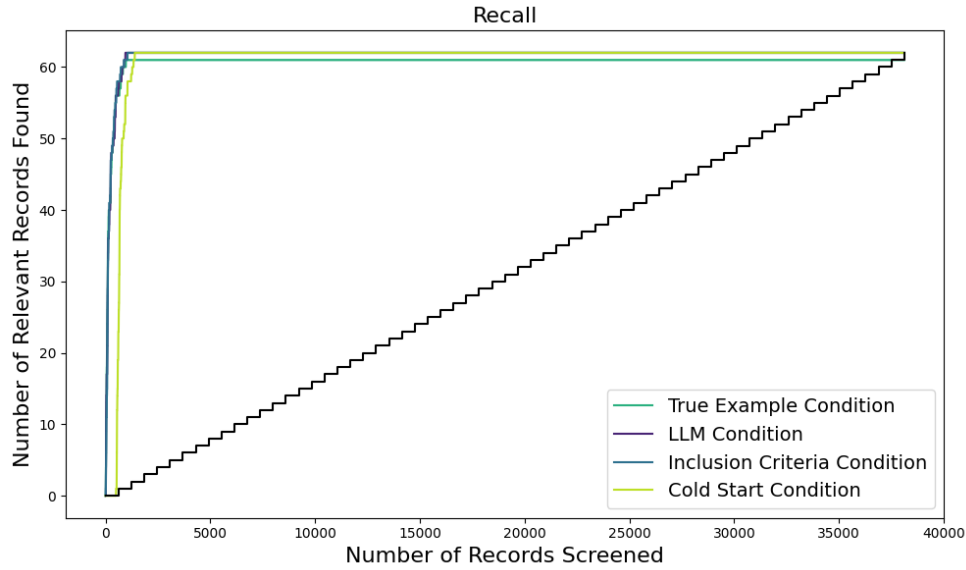
We are interested in comparing the starting performance of the four conditions. To that end, figure 2 presents four recall plots, one for each condition. Each plot contains simulation runs for two datasets: Appelneller-Herzog (2019) and Brouwer (2019). These datasets and the specific plots below were selected because they represent the overall findings of the simulation study well, and thus serve as appropriate didactic examples to illustrate the differences in starting performance between the four conditions.

We can see that there are clear differences in starting performance between the cold start condition and the other three conditions. In the cold start condition, the recall curves are generally flat, indicating poor starting performance (mean number of papers found = 10). In the LLM and criteria-only conditions, the recall curves are generally

⁴For example, "rodents" may be expanded to include hyponyms such as "mice" and "rats", and "Wilson's disease" to include synonyms such as "Progressive Lenticular Degeneration"



(a) Recall plot for the Appelneller-Herzog (2019) dataset



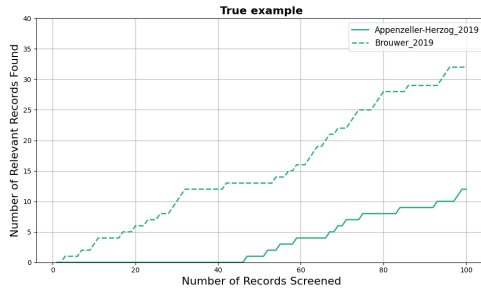
(b) Recall plot for the Brouwer (2019) dataset

Figure 2: Aggregate recall plots for two datasets across all four conditions. The x-axis shows the number of papers screened (0–100) and the y-axis shows the number of relevant papers found. Each line shows one run of each condition.

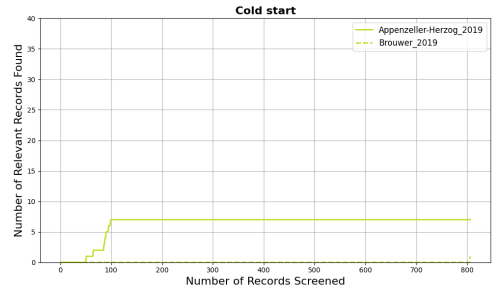
steeper, indicating better starting performance (mean number of papers found = 22 and 21, respectively). Finally, in the true example condition, performance was generally on par with the LLM and criteria-only conditions (mean number of papers found = 17). However, we chose also show one run with poor starting performance to illustrate the fact that the example at hand may not be a representative example of the relevant papers in the systematic review, which may lead to suboptimal starting performance.

Indeed, there is considerable variability in starting performance across runs within all the conditions. As such, these plots should not be taken as representative of the performance of each condition in general. The following sections will present aggregate recall plots and boxplots of starting performance across all runs to provide a more comprehensive overview of the performance of each condition.

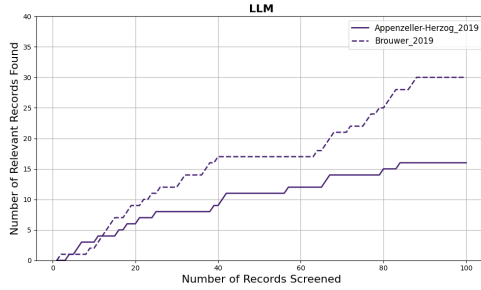
3.2 Comparing starting performance: recall curves for each condition



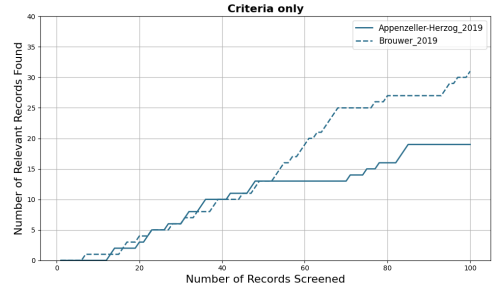
(a) Recall true examples condition



(b) Recall cold start condition

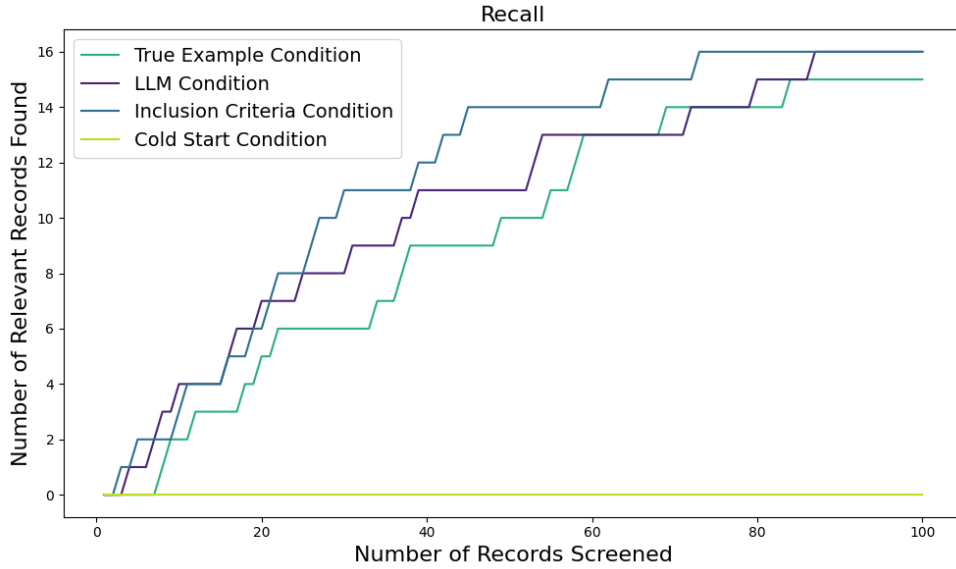


(c) Recall LLM condition

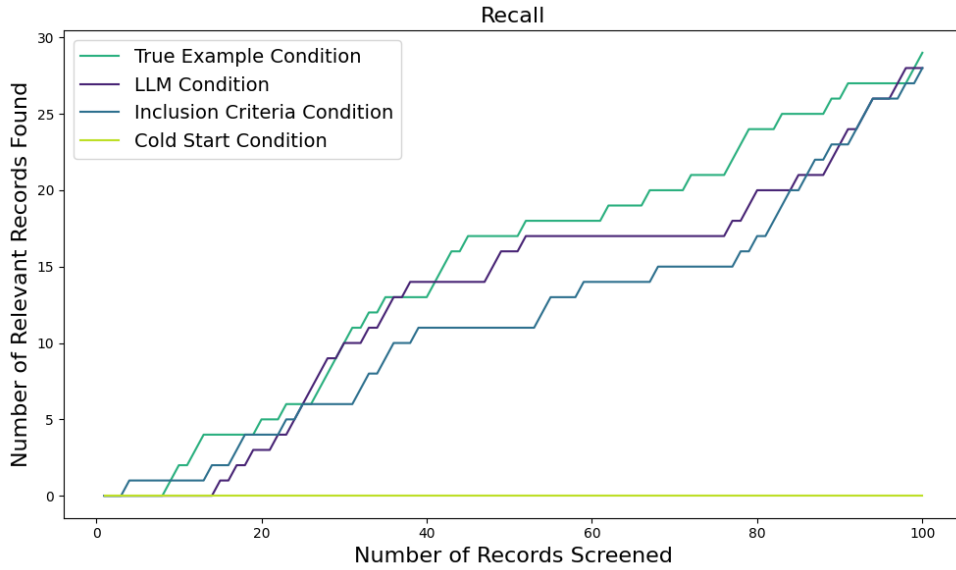


(d) Recall criteria-only condition

Figure 4: Four recall plots of single runs of two datasets per condition (as didactic examples)



(a) Recall plot for the Appelneller-Herzog (2019) dataset

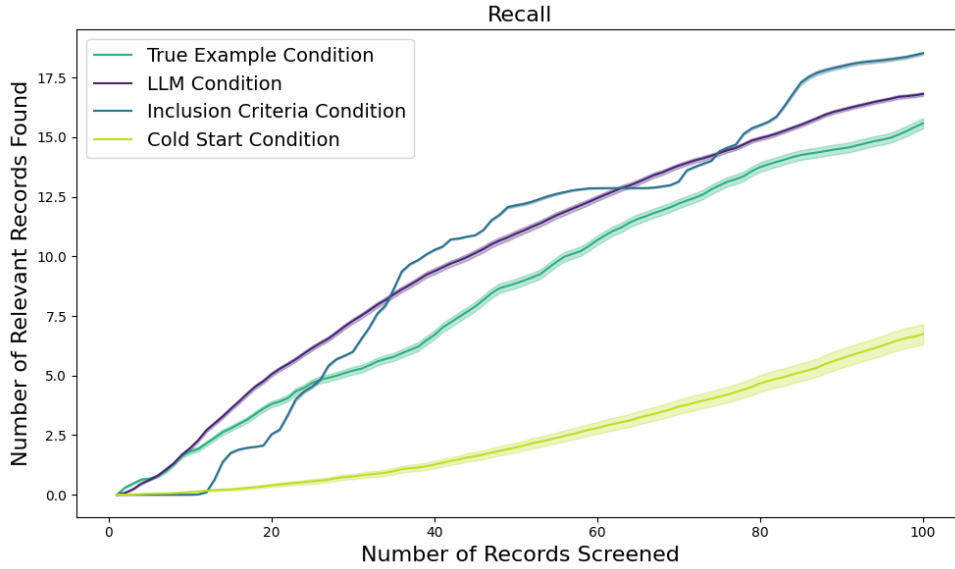


(b) Recall plot for the Brouwer (2019) dataset

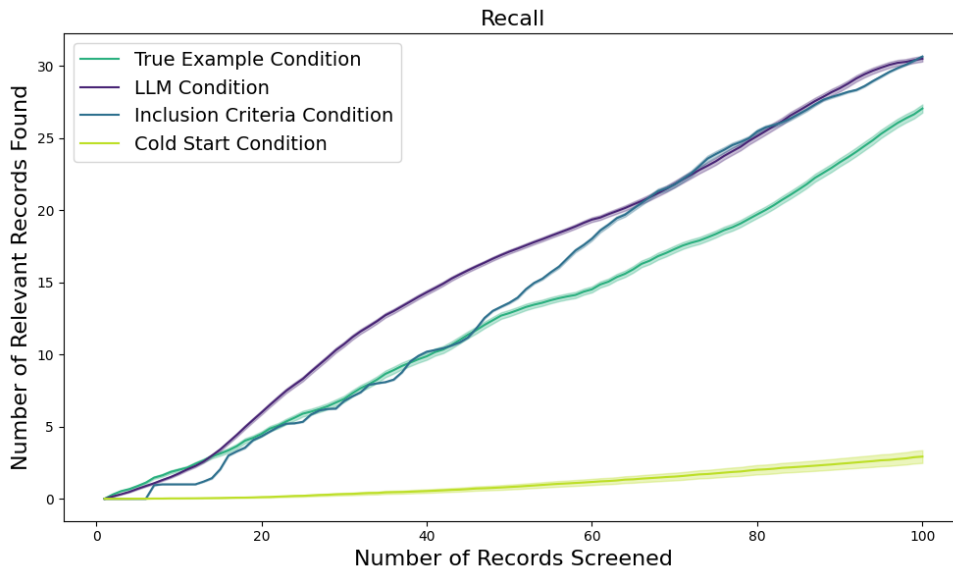
Figure 3: Aggregate recall plots for two datasets across all four conditions. The x-axis shows the number of papers screened (0–100) and the y-axis shows the number of relevant papers found. Each line shows one run of each condition.

3.3 Aggregate recall plots with line for each condition, for two datasets

The recall plots shown above are just examples of single runs. By simulating many many runs, we can obtain a more reliable estimate of the average performance of each condition. (i.e., do statistical analysis / significance testing).



(a) Recall plot for the Appelneller-Herzog (2019) dataset



(b) Recall plot for the Brouwer (2019) dataset

3.4 Boxplots of starting performance across all runs

3.5 Few sentences describing the results of the statistical analysis

One alinea on reporting the statistical significance of the three contrasts: (1) LLM vs. cold-start, (2) criteria-only vs. cold-start, and (3) LLM vs. criteria-only.

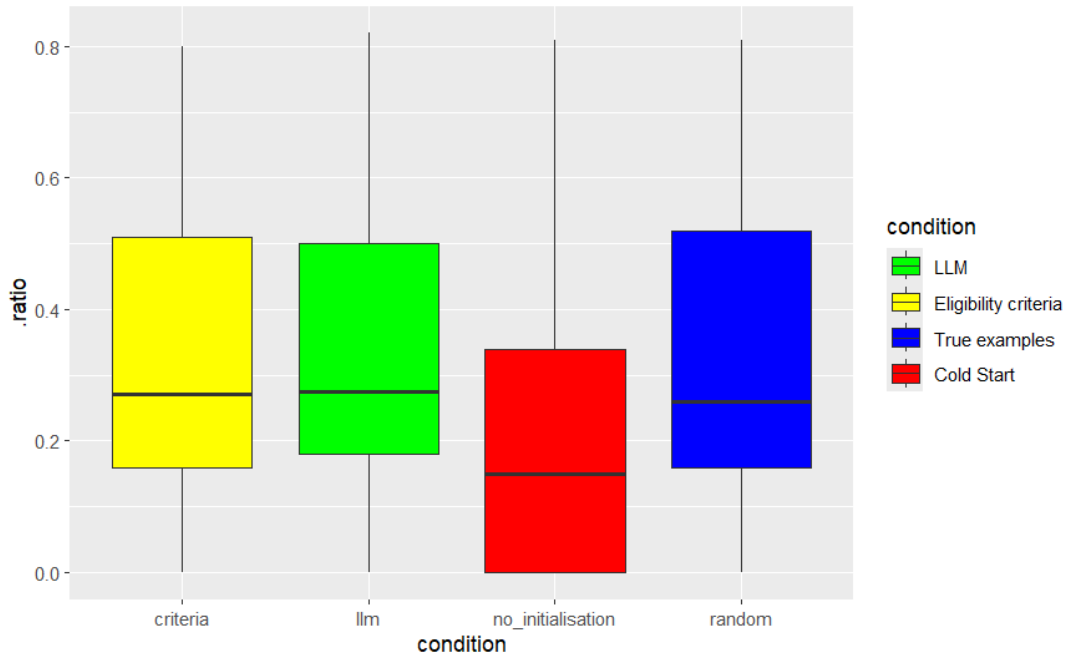


Figure 6: Example boxplot

3.6 Additional analysis on using eligibility criteria as prior

One alinea reporting the results of manually manipulating / enriching the eligibility criteria.

4 Conclusion

The conclusion is that the cold start problem can be overcome by using the eligibility criteria of a systematic review to initialize the active learning model, and that LLM-generated examples do not improve starting performance beyond that achieved through initialisation using the systematic review’s eligibility criteria criteria directly.

5 Discussion

A limiting factor in the practical application of AI-assisted systematic reviews is the cold start problem. That is, the inability of the AI-models in question to prioritize relevant papers for screening when it has yet to be trained on any examples of relevant papers limits the starting performance of AI-assisted systematic reviews in situations where no prior examples of relevant papers are available. This is especially relevant to crisis situations such as COVID-19, where there may not be relevant prior literature and timely evidence synthesis is crucial.

1. A limiting factor in the application of AI-assisted systematic reviews is the cold start problem. This is especially relevant to crisis situations such as COVID-19, where there may not be relevant prior literature and timely evidence synthesis is crucial.

2. Prior research has shown that the cold start problem can be overcome by providing as little as one true pair example of a relevant paper and an irrelevant paper to the active learning model before screening begins (Ferdinands et al. 2023; Teijema et al. 2025). However, in some cases, there may be no prior examples of relevant papers available to initialize the model with. Moreover, the true examples a user has available may not be representative of the relevant papers in the systematic review, which may lead to suboptimal starting performance (i.e., bias / anchor effect – ask Sixu).
3. Currently, the cold start problem is typically addressed by consulting a domain expert to provide true examples of relevant papers to initialize the model with (Mikkola et al. 2024).
4. Some limitations of the current study include:
 - (a) Only a limited number of historical datasets were used. SYNERGY 2.0 coming soon!
 - (b) The of historical datasets. Future studies may consider addressing the cold start problem in a new systematic review.
 - (c) ...
5. Practical implications of the current study include:
 - (a) If no prior examples of relevant papers are available, the eligibility criteria of a systematic review can be used to initialize the active learning model and overcome the cold start problem.
 - (b) Don't use LLMs. LLMs add very little at most and there are many disadvantages to their use.
 - (c) If one true example of a relevant paper is available, still use the eligibility criteria to initialize the model. This way your true paper can be used as a test for completeness of your screening (i.e., stopping rule is reached when the true example is found – cite Tale & Josien).
 - (d) ...

A SYNERGY metadata

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Table 1: Datasets overview

Nr	Dataset	Topic(s)	Records	Included	%
1	Appenzeller-Herzog_2019	Medicine	2873	26	0.9
2	Bos_2018	Medicine	4878	10	0.2
3	Brouwer_2019	Psychology, Medicine	38114	62	0.2
4	Chou_2003	Medicine	1908	15	0.8
5	Donners_2021	Medicine	258	15	5.8
6	Hall_2012	Computer science	8793	104	1.2
7	Leenaars_2019	Psychology, Chemistry, Medicine	5812	17	0.3
8	Leenaars_2020	Medicine	7216	583	8.1
9	Meijboom_2021	Medicine	882	37	4.2
10	Menon_2022	Medicine	975	74	7.6
11	Moran_2021	Biology, Medicine	5214	111	2.1
12	Muthu_2021	Medicine	2719	336	12.4
13	Nelson_2002	Medicine	366	80	21.9
14	Oud_2018	Psychology, Medicine	952	20	2.1
15	Radjenovic_2013	Computer science	5935	48	0.8
16	Sep_2021	Psychology	271	40	14.8
17	Smid_2020	Computer science, Mathematics	2627	27	1.0
18	van_de_Schoot_2018	Psychology, Medicine	4544	38	0.8
19	van_der_Valk_2021	Medicine, Psychology	725	89	12.3
20	van_der_Waal_2022	Medicine	1970	33	1.7
21	van_Dis_2020	Psychology, Medicine	9128	72	0.8
22	Walker_2018	Biology, Medicine	48375	762	1.6
23	Wassenaar_2017	Medicine, Biology, Chemistry	7668	111	1.4
24	Wolters_2018	Medicine	4280	19	0.4

Table 2: Please note that two of the datasets included in the original SYNERGY dataset were excluded entirely due to data quality issues: Chou (2003) and Jeyaraman (2020). For one dataset (Moran, 2021), an updated version was used due to data quality issues in the original version.